Ch 6.2: Shrinkage - The Lasso

Lecture 19 - CMSE 381

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Weds, Oct 18, 2023

Announcements

Last time:

- Ridge Regression

This time:

- The Lasso

Announcements:

- HW # 5 due tonight
- Friday Review Class
- Monday No class
- Weds Exam #2

Section 1

Last time

Goal

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

↳ Last time .... Ridge regression

Ridge regression

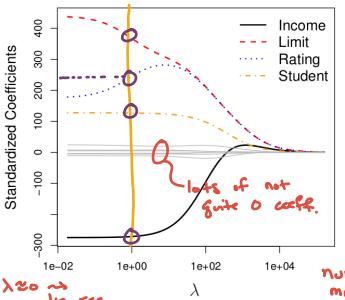
Before:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

After:

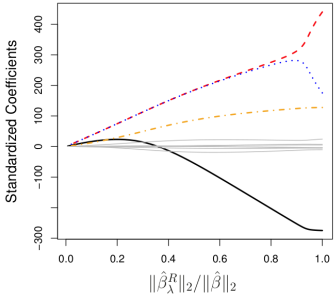
$\lambda \geq 0$ hyper param.

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\underbrace{\sum_{j=1}^{p}\beta_j^2}_{L_2 \text{ norm}}$$



$\lambda \approx 0 \rightarrow$ lin reg.

null model (predict avg {y} every time)



Scale equivariance (or lack thereof)

Scale equivariant: Multiplying a variable by c (cX_i) just returns a coefficient multiplied by $1/c$ ($1/c\beta_i$)

ex) Income variable $c = 1000$

in 1000's of \$'s or \$'s

$\beta_j \leftarrow \begin{cases} 54 \\ 123 \end{cases}$ $\quad \left.\begin{matrix} 54{,}000 \\ 123{,}000 \end{matrix}\right\} \rightarrow \frac{1}{1000}\beta_j$

Solution: standardize predictors

$$\widetilde{x}_{ij} = \underbrace{\frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}}_{\text{divide by std dev of the column}}$$

- Least squares is scale equivariant
- Ridge regression is not

Section 2

The Lasso

Same goal as before

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

↳ Ridge regression tends to shrink the coeff to 0, but not actually get to 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

The lasso

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

Ridge:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = \boxed{RSS + \lambda\sum_{j=1}^{p}\beta_j^2}$$

L_2

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = \boxed{RSS + \lambda\sum_{j=1}^{p}|\beta_j|}$$

L_1 norm

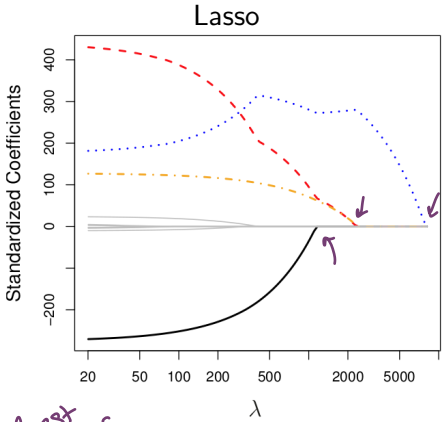
hyperparam $\lambda \geq 0$

Subsets with lasso

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

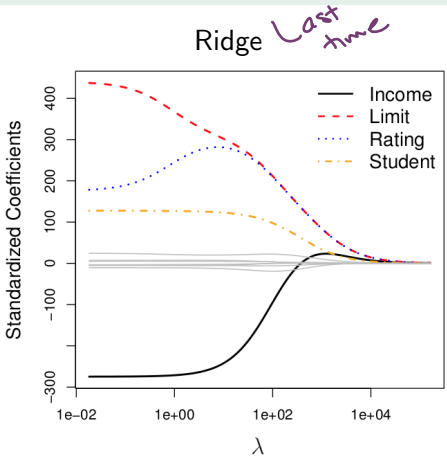
- Result of L_1 penalty is that it forces coeff to actually be 0
- Can interpret as something similar to variable | subset selection
  - ↳ just take variables learned w/ $\hat{\beta}_i > 0$
- Yields "sparse models" (uses fewer variables)
- Like ridge, selecting good λ is important

An example on `Credit` data set

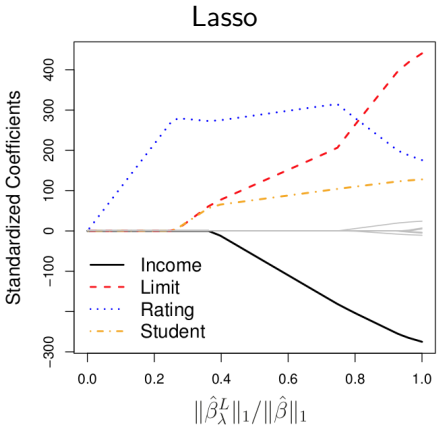


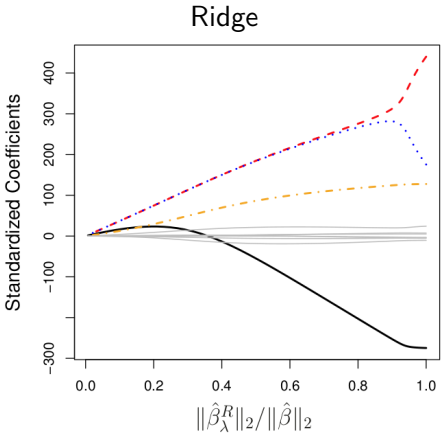
least squares
Linear reg $\lambda=0$

null model



More example on `Credit` data set





Scale equivavariance (or lack thereof)

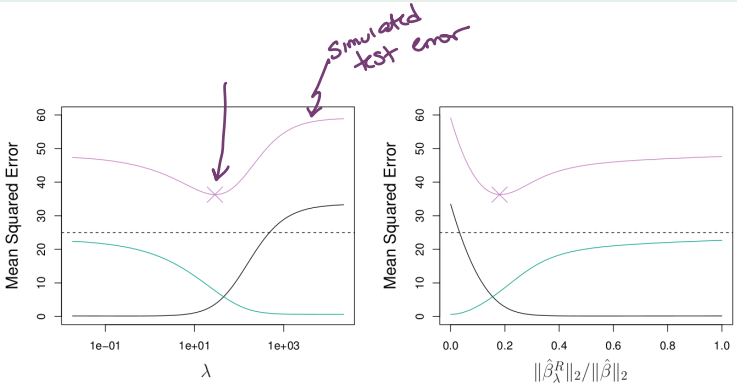
Scale equivariant: Multiplying a variable by c just returns a coefficient multiplied by $1/c$

Least squares **is** scale equivariant.
Ridge/Lasso **are very much not**.

Solution: standardize predictors

$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

Bias-Variance tradeoff

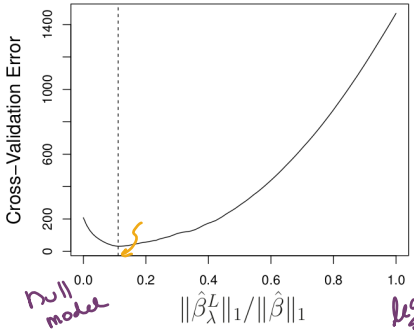


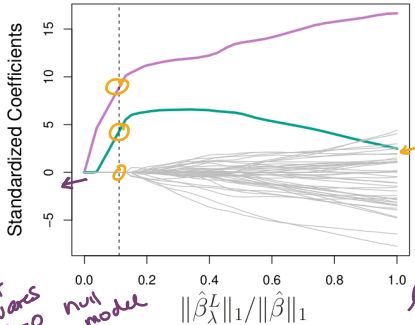
Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

Using Cross-Validation to find λ

- Choose a grid of λ values
- Compute the (k-fold) cross-validation error for each value of λ
- Select the tuning parameter value λ for which the CV error is smallest.
- The model is re-fit using all of the available observations and the selected value of the tuning parameter.

10-fold CV choice of λ for lasso and simulated data





In reg doesn't show green as contributing much

* Data gen w/ $p=45$, $n=50$ data pts, only used 2 variables to get y

Coding example

Section 3

Optimization Formulation

Another formulation for Ridge Regression

Find β to minimize:

$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

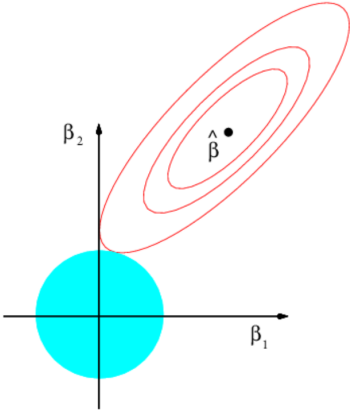
Find β to minimize

$$RSS$$

subject to

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

Visualization using disks



Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to

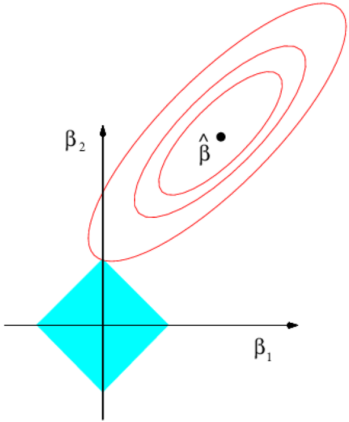
$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

What about ℓ_1?

$$\|\beta\|_1 = \sum |\beta_i|$$

What does the set of points (β_1, β_2) for which $\|(\beta_1, \beta_2)\|_1 \leq s$ look like?

Same game for Lasso



Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to

$$\sum_{j=1}^{p} |\beta_j| \leq s$$

Same game for subset selection

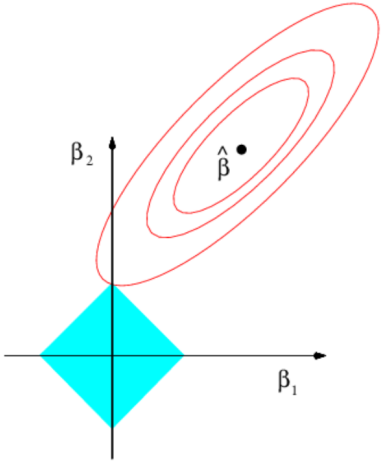
Find β to minimize

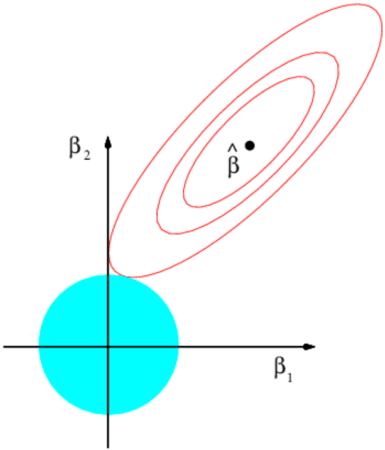
$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to

$$\sum_{j=1}^{p} \mathrm{I}(\beta_j \neq 0) \leq s$$

Using this visual to understand why lasso gets us zero values





TL;DR - Original forumlation

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

Ridge:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

TL;DR

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

Least Squares:

No constraints

Ridge:

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$

Next time

12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2	
13	Wed	Oct 4	k-fold CV	5.1.3	
14	Fri	Oct 6	More k-fold CV,	5.1.4-5	
15	Mon	Oct 9	k-fold CV for classification	5.1.5	HW #4 Due
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2	
17	Fri	Oct 13	Subset selection	6.1	
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1	
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2	
	Fri	Oct 20	**Review**		
	Mon	Oct 23	No class - Fall break		
	Wed	Oct 25	**Midterm #2**		
20	Fri	Oct 27	Dimension Reduction	6.3	